Gamson's Law or a Formateur Premium?

Political Parties as Multi-Agent Reinforcement Learning Agents
Bargaining over Coalition Cabinets

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Code, data and trained agents: https://github.com/bot2048/cabinet-bargaining-marl

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Abstract

Coalition cabinets divide ministries almost exactly in proportion to the seats each partner contributes (Gamson's law), whereas the leading non-cooperative theory of legislative bargaining (Baron–Ferejohn) predicts that the formateur, the party that proposes the cabinet, takes a large premium. We ask which of the two allocation rules self-interested parties arrive at when they learn to bargain. Parties are modelled as reinforcement-learning agents in a configurable game of government formation, rewarded only by their own office and policy payoffs. A single permutation-equivariant policy is trained by proximal policy optimisation and checked against exact equilibria, which we derive for the proposal protocol and compute for the demand protocol. Three results emerge. (i) Under proposal bargaining the learned formateur keeps a premium (seat-controlled coefficient +0.249 in the base game) that no institution we add removes: private valuations and ministerial policy remove it from the *number* of ministries but not from their value or from policy control. (ii) Under demand bargaining the premium disappears (pre-registered test, exact permutation $p = 5 \times 10^{-6}$), but learned shares follow *voting weights*, not seats: in a 45/35/20 parliament every pair splits 50/50, as the exact equilibrium predicts and Gamson's law does not. (iii) In 195 real coalition cabinets the prime minister's party has no premium relative to seats (0.011), a small one relative to voting weights (+0.074), and partners with equal voting weight still tend to divide portfolios by seats (slope 0.84, 95% CI 0.44–1.07). Learning parties thus favour the formateur exactly as far as the rules give it proposal power and otherwise split by voting weight; real cabinets follow seats even where seats and voting weights disagree, which points to explanations outside these bargaining models.

Keywords: coalition government; portfolio allocation; Gamson's law; Baron–Ferejohn bargaining; demand bargaining; multi-agent reinforcement learning; exploitability.

1 Introduction

When an election gives no party a majority, parties bargain over who governs and over the division of ministries. Ministries are the main currency of office, and their division is one of the most robust regularities in comparative politics: cabinet portfolios are shared in proportion to the seats each coalition partner brings, with a slight bias toward small partners (Gamson, 1961; Browne and Franklin, 1973; Warwick and Druckman, 2006). This regularity, Gamson's law, sits uneasily with non-cooperative bargaining theory. In the closed-rule model of Baron and Ferejohn (1989), a recognised proposer buys the cheapest majority at its partners' continuation values and keeps the rest, so the *formateur* should take a large premium (Figure 1). Laboratory evidence finds a proposer advantage that is smaller than predicted (Fréchette et al., 2003, 2005), and measuring party strength by voting weights instead of seats reveals a formateur bonus in real

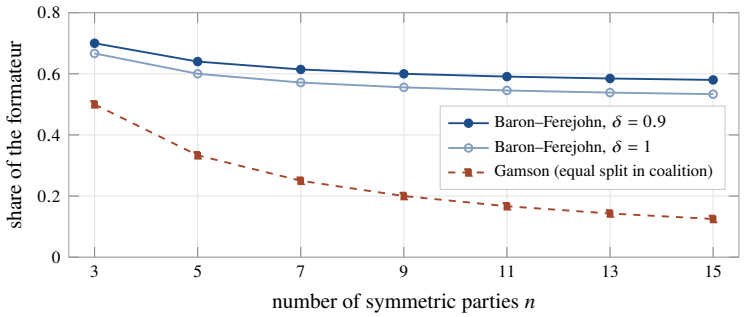


Figure 1: The puzzle in the symmetric case: the formateur's equilibrium share under Baron–Ferejohn (solid) and the equal split prescribed by Gamson's law (dashed), as the number of parties grows.

data (Ansolabehere et al., 2005). Which allocation rule bargaining parties actually arrive at, and under which institutions, therefore remains open.

We study the question computationally. Parties are reinforcement-learning (RL) agents in a configurable game of government formation that nests the Baron–Ferejohn game exactly and adds, one switch at a time, the institutional features that have been proposed as explanations of Gamson's law: repeated terms, exits, the recognition rule, heterogeneous and private valuations of ministries, ministerial policy and a demand-based bargaining protocol. Agents are rewarded only by their own payoffs, so any proportionality must emerge; nothing in the reward refers to fairness. We read learned behaviour through two lenses: as an approximate equilibrium solver, checked against exact equilibria and by best-response (exploitability) tests, and as a model of adaptive, boundedly rational parties.

Contributions.

1. An open environment and learner for coalition bargaining: one permutation-equivariant attention policy with masked autoregressive proposal heads serves any number of parties and ministries, and is trained by PPO on each agent's own decision sequence (Sections 3 and 4).
2. Exact benchmarks: the Baron–Ferejohn solution, embedded in the environment, and a verified backward-induction solver for our demand protocol, whose equilibrium is not covered by published results (Section 3.2). The equal split it predicts in 45/35/20 follows from the protocol; the learning result is that agents find it.
3. Evidence that matching equilibrium *outcomes* does not imply equilibrium *play*: learned Baron–Ferejohn agents reproduce the closed-form shares while leaving a profitable formateur deviation untried (Section 5.1).
4. A mechanism map of the learned formateur premium in three currencies: number of ministries, the formateur's own valuation, and policy-setting ministries (Section 5.2).
5. Pre-registered tests of the bargaining protocol, including a decisive test of whether learned shares follow seats or voting weights (Section 5.3).
6. An out-of-sample confrontation with 195 real coalition cabinets, including the agents' own cabinets in the same parliaments and a direct test of seats against voting weights in real data (Section 5.4).

Answer in brief. Learning parties do not produce seat-proportionality as such. They favour the formateur exactly as far as the protocol gives it proposal power, and otherwise divide by voting weight; information and policy motives change the currency of the premium, not its existence. Real cabinets follow seats even where voting weights disagree.

2 Related work

Gamson's law. Since Browne and Franklin (1973), regressions of portfolio shares on within-coalition seat shares have found slopes near one and very tight fits, also when ministries are weighted by salience (Warwick and Druckman, 2001, 2006). Small partners receive slightly more than proportional shares. Ansolabehere et al. (2005) show that a formateur advantage appears when strength is measured by minimal integer voting weights. Explanations of proportionality include pre-electoral pacts (Carroll and Cox, 2007), uncertainty and complexity (Falcó-Gimeno and Indridason, 2013), commitment to a coalition before the division (Kamm and Siegenthaler, 2024) and within-party bargaining (Cox, 2021).

Bargaining theory. In the closed-rule model (Baron and Ferejohn, 1989) the proposer extracts a premium whose size depends on patience and the number of parties; stationary equilibrium payoffs are unique (Eraslan, 2002). With weighted voting, expected payoffs follow voting weights and recognition probabilities while the realised proposer premium remains (Snyder et al., 2005); recognition rules can support almost any expected division (Kalandrakis, 2006). Policy can be traded for office (Baron, 1991; Laver and Shepsle, 1996). Demand bargaining (Morelli, 1999) predicts no proposer premium and shares proportional to voting weights for specific move orders; Montero and Vidal-Puga (2007) give a counterexample to the general claim and Montero and Vidal-Puga (2011) restore proportionality when parties move in decreasing order of weight. Our protocol fixes neither order, so we solve it directly.

Learning and experiments. Laboratory subjects give proposers less than the theory predicts (Fréchette et al., 2003) and react less to the protocol than theory suggests (Fréchette et al., 2005). Learning dynamics can settle on outcomes that are not subgame perfect, because responses that are rarely tested on the path of play are never corrected (Gale et al., 1995; Fudenberg and Levine, 1993, 1998). Deep multi-agent RL has been used for negotiation (Lewis et al., 2017), team formation in weighted voting games (Bachrach et al., 2020), economic policy design (Zheng et al., 2022) and pricing (Calvano et al., 2020). To our knowledge, no prior study combines the three elements used here: learning agents in a configurable cabinet-bargaining game, validation against protocol-specific exact equilibria including best-response tests, and confrontation with real cabinet allocations. Our contribution is this combination, not any single component.

3 Model

3.1 The cabinet bargaining game

A parliament has parties $\mathcal{N} = \{1, \ldots, n\}$ with seats w_i, total W and quota $q = \lfloor W/2 \rfloor + 1$; a coalition C is winning if $w(C) = \sum_{i \in C} w_i \geq q$. There are m ministries with salience σ_k and, optionally, party-specific valuations $\nu_{ik} = (1 - \xi)\sigma_k + \xi\, \epsilon_{ik}$ with $\epsilon_{i\cdot}$ drawn from a Dirichlet distribution. Each period without a cabinet is a formation round (Figure 2).

Proposal protocol. A formateur f is drawn from the recognition rule ρ (uniform or proportional to seats). It proposes a winning coalition $C \ni f$ and an allocation $\mu : \{1..m\} \to C$ (or, with a divisible surplus, shares $p \in \Delta(C)$). The cabinet forms if every partner in C accepts; otherwise the period is lost and the next round is discounted by δ.

Demand protocol. A first mover is recognised by ρ; the others move in a uniformly random order. Each mover either demands a share of a divisible surplus on a 1% grid, or *closes*: it forms a cabinet with the cheapest winning set of earlier demands whose sum is at most one (ties: fewer members first) and takes the residual. If nobody closes, the round fails. This differs from Morelli (1999), where the first mover chooses the order of moves.

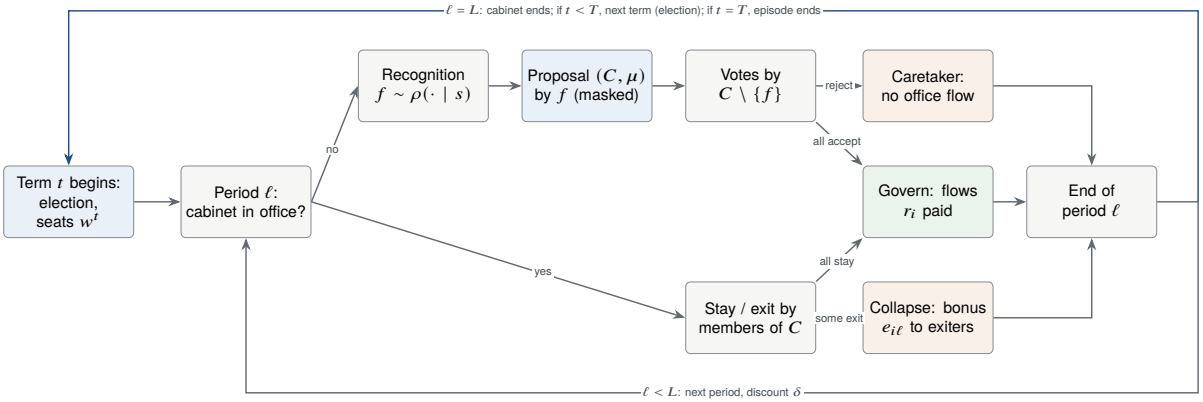


Figure 2: One period of the environment under the proposal protocol. Without a cabinet, a formateur is recognised and proposes a coalition and an allocation, which forms the cabinet if every named partner accepts; otherwise a caretaker period passes. A governing cabinet pays per-period flows; with exits, members may leave and collapse it.

Payoffs. Each party i has an ideal point $z_i \in [0,1]^d$ ($d = 2$). The first d ministries are *lead* ministries, one per policy dimension. In every period, governed or not, policy is $y_h = z_{\mu(h),h}$ (the ideal point of the party holding lead ministry h) while a cabinet governs, and the status quo $y_h = y^0 = 0.5$ in a caretaker period. Party i's per-period reward is

$$r_i = \underbrace{\textstyle\sum_{k:\mu(k)=i} \nu_{ik}}_{\text{office (members only)}} - \lambda \, \tfrac{1}{d} \textstyle\sum_{h=1}^{d} |y_h - z_{ih}|, \tag{1}$$

so members and excluded parties alike bear the policy loss of the policy in force, and in a caretaker period nobody receives office. A member that exits receives its private exit bonus instead and the cabinet collapses. Rewards are individual; there is no fairness term. Parameter values are in Section D.

Mechanisms. Each candidate explanation is a switch: repeated terms with elections (M1), the demand protocol (M2), private valuations (M4, with heterogeneous values M5), exits (M8), ministerial policy (M9) and seat-proportional recognition (M10). The *base* game has indivisible ministries and every mechanism off; E* has all of them on. Level E0 is the symmetric Baron–Ferejohn game with a divisible surplus. The game is a partially observable stochastic game; private valuations and ideal points are observed only by their owner.

Measurement. On every installed cabinet c with portfolio-holding members we estimate the standard regression

$$p_{ic} = \alpha + \beta \, s_{ic} + \gamma \, F_{ic} + \varepsilon_{ic}, \tag{2}$$

with p_{ic} the share of ministries, s_{ic} the within-coalition seat share and F_{ic} the formateur dummy (for real cabinets, the prime minister's party). Gamson's law is $(\alpha, \beta, \gamma) = (0, 1, 0)$; γ is the seat-controlled formateur premium and our headline statistic. Because the premium may move to other currencies, we estimate (2) also with each party's share of the value *it* places on ministries (value premium) and its share of policy-setting ministries (lead premium), and with Banzhaf voting power or minimal integer voting weights in place of seats. With m indivisible ministries exact proportionality is usually infeasible, so we also report statistics relative to the most proportional feasible allocation of the same cabinet,

$$p^{\min}(C) = \arg \min_{\mu:\{1..m\}\to C} \textstyle\sum_{i \in C} |p_i(\mu) - s_i|,$$

computed by Hamilton's largest-remainder rule for equal salience, the Gamson distance $D_G = \frac{1}{2}\sum_{i \in C} |p_i - s_i|$, its floor-adjusted version $\tilde{D}_G = \mathbb{E}[D_G(p) - D_G(p^{\min})]$, and the floor-adjusted formateur premium $\tilde{\Phi} = \mathbb{E}[p_f - p_f^{\min}]$.

3.2 Theoretical benchmarks

Proposition 1 (Symmetric Baron–Ferejohn benchmark)**.** *In level E0 with n odd, equal seats, uniform recognition and discount δ, the stationary subgame-perfect equilibrium in stage-undominated voting strategies has the first proposal accepted; the formateur offers δ/n to $(n-1)/2$ partners and keeps $x_f^{\mathrm{BF}} = 1 - \delta(n-1)/(2n)$.*

The environment embeds the Baron–Ferejohn game exactly when a cabinet governs forever and shares are paid as per-period flows; the proof is in Section A. For weighted parliaments we compute stationary equilibria with a homotopy root finder certified by a linear-programming residual check (`cabinet/bf_solver.py`).

Exact equilibrium of our demand protocol. Published results do not cover our variant, so we compute the subgame-perfect equilibrium of the game as implemented, with demands on a finite grid; the solution is exact for that finite game, not for every demand-bargaining model. We compute it by backward induction over every history of grid demands and every move order, with failure payoffs δV_i iterated to a fixed point in the ex-ante values V (`cabinet/demand_solver.py`, Section B). It agrees with an independently written brute-force solver and with the environment's closing rule (Section B). The results do not depend on the tie-breaking rule, the grid (1% and 2%; 2.5% and 4% for five parties) or patience ($\delta \in \{0.5, 0.9, 0.95\}$):

- three equal parties: the first two movers split 0.50/0.50; five equal parties: the first three movers take about one third each;
- 45/35/20 seats: *every* pair is winning, and whichever pair forms splits 0.50/0.50, whatever its seats;
- 40/30/25/5 seats: the 5-seat party is a dummy (never pivotal) and gets nothing; two of the large parties split 0.50/0.50.

The undercutting logic is simple: under the cheapest-set rule an earlier demand above the closer's residual is replaced by a cheaper later demand, so demands are competed down to the split of the minimal winning coalition. The equilibrium is proportional to minimal integer voting weights, here $(1,1,1)$ and $(1,1,1,0)$, not to seats. Gamson's law predicts 0.56/0.44 to 0.69/0.31 for the 45/35/20 pairs.

4 Learning framework

Policy. All parties share one network (`cabinet/policy.py`). The observation of a party is a *set* of party tokens (seat share, recognition probability, roles, current offers, the last cabinets), a set of ministry tokens (salience, own valuation, policy dimension), pair features and a context vector (phase, quota, δ, λ, expected terms). Two attention blocks make the policy permutation-equivariant, so it applies to any n and m and treats relabelled parties identically. Proposals are decoded autoregressively with masks that keep every proposal feasible: include or exclude each party in order of seats, with the quota enforced, then point every ministry to a coalition member; for a divisible surplus, a 1% grid offer per partner. Voters, exiting members and demanders have their own heads; the demand head has 101 demand bins and one *close* action. An optional GRU core carries memory across decisions.

Table 1: Benchmark E0 (symmetric closed-rule Baron–Ferejohn game, divisible surplus; 10 seeds, 4000 evaluation episodes each). Gap: learned minus closed-form formateur share, with a 95% t-interval over seeds. Threshold: offer at which a learned voter accepts with probability 1/2. BR gain: payoff a formateur would gain per formation by best-responding to the frozen learned voters (mean over seeds).

n	δ	x_f^{BF}	learned	gap [95% CI]	threshold	δ/n	BR gain
3	0.8	0.733	0.733	−0.001 [−0.008, +0.007]	0.205	0.267	0.013
3	0.9	0.700	0.706	+0.006 [−0.008, +0.020]	0.257	0.300	0.005
3	0.95	0.683	0.694	+0.010 [−0.004, +0.025]	0.273	0.317	0.003
5	0.8	0.680	0.689	+0.009 [−0.039, +0.057]	0.079	0.160	0.056
5	0.9	0.640	0.662	+0.022 [−0.017, +0.061]	0.122	0.180	0.019
5	0.95	0.620	0.650	+0.030 [−0.003, +0.064]	0.134	0.190	0.014
7	0.8	0.657	0.601	−0.056 [−0.079, −0.033]	0.025	0.114	0.153
7	0.9	0.614	0.600	−0.014 [−0.036, +0.008]	0.055	0.129	0.081
7	0.95	0.593	0.598	+0.005 [−0.016, +0.026]	0.064	0.136	0.058
mean signed gap +0.001; mean absolute gap 0.017; 6/9 cells within 0.02							

Training. PPO (Schulman et al., 2017) with generalised advantage estimation (Schulman et al., 2016) is applied to each agent's own sequence of decisions, a semi-Markov decision process (`cabinet/ppo.py`). Between an agent's decisions at periods t and $t+\Delta$ it keeps receiving rewards, which enter its transition as $R_t = \sum_{\tau=0}^{\Delta-1} \delta^\tau r_{i,t+\tau}$ with discount δ^Δ on the next value, so the advantage uses $R_t + \delta^\Delta V(o_{t+\Delta}) - V(o_t)$; decisions within a period ($\Delta = 0$) are undiscounted. Advantages are normalised per decision type, and the vote and exit heads get five times the entropy bonus so that responders keep learning. Each iteration collects 8192 decisions; runs last 500–2000 iterations. Variants: MAPPO with a centralised critic that sees private types (Yu et al., 2022), fictitious self-play against a pool of past checkpoints (Heinrich and Silver, 2016), symmetric PSRO (Lanctot et al., 2017) and the GRU memory. Hyper-parameters (Section C) were tuned on E0 only.

Exploitability. For a frozen profile we train a best response by PPO for one randomly drawn seat while every other seat plays the profile, from the profile's weights and from scratch, and report the larger per-period gain over 3000 evaluation episodes. Scripted Gamson and Baron–Ferejohn-heuristic profiles serve as positive controls. A learned best response bounds exploitability from below, so a small gain means only that no exploit was found at the budget.

Statistics. Training seeds are the independent units. Each seed's coefficients come from its own evaluation cabinets; we report means with 95% intervals over seeds. Before the final runs finished we committed a pre-registration (`docs/PREREGISTRATION.md`) fixing three confirmatory tests with Holm correction and four descriptive predictions; everything else is exploratory. For real data we use country-clustered and wild-cluster bootstraps (Cameron et al., 2008).

5 Results

5.1 Benchmark: learned outcomes match Baron–Ferejohn, learned play does not

In E0, learned formateur shares follow the closed form of Proposition 1 (Figure 3, Table 1): the mean absolute gap over the nine cells is 0.017, the mean signed gap +0.001, and 6/9 cells are within 0.02; shares fall with patience δ as theory predicts for $n \leq 5$. Between 90% and 100% of cabinets are minimal winning, and 2.2% of rounds fail on average. The largest gap is at $n = 7$, $\delta = 0.8$, where the learned share stays at about 0.60 for every δ.

Matching outcomes is not equilibrium play. Probing learned voters with synthetic offers shows acceptance thresholds well below the reservation value δ/n for $n \geq 5$ (Table 1). At the

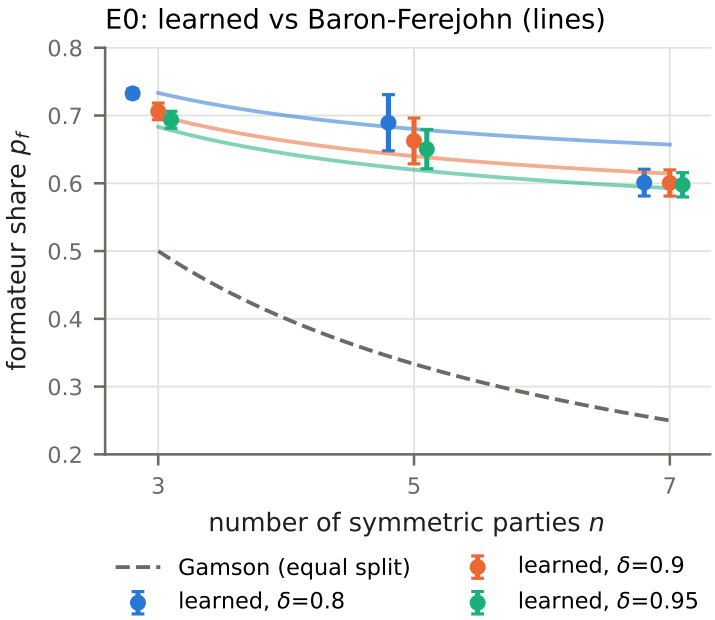


Figure 3: Benchmark E0: learned formateur share (mean and 95% CI over 10 seeds) against the closed form of Proposition 1; dashed: Gamson's equal split.

offers formateurs actually make, the probe agrees with on-path behaviour (acceptance near one, failure near zero), but against the frozen voters a formateur best response would gain up to 0.153 per formation at $n = 7$ and 0.056 at $n = 5$, by offering far less (`scripts/bf_best_response.py`). Formateurs never try such offers, so voters are never taught to reject them: the self-confirming pattern of learning in games (Fudenberg and Levine, 1993; Gale et al., 1995). As in the laboratory (Fréchette et al., 2003), the proposer ends up with less than the theory gives it at large n. A best response trained by PPO against the E0 profile gains 0.040 per period (20% of a seat's payoff), confirming that the exploitability test detects such gaps.

5.2 Proposal bargaining: the premium changes currency

Table 2 and Figure 4 report the formateur coefficient as mechanisms are added to the base game or removed from E*. The base game keeps a substantial premium ($\hat{\gamma} = +0.249$, $+0.295$ with four times the training budget), and it emerges with training rather than being present from the start (Figure 6 in Section E). Repeated terms, even with patient agents and memory, exits, the recognition rule and heterogeneous public valuations reduce it at most partially. Every learner (IPPO, MAPPO, fictitious self-play, PSRO, GRU) reproduces the pattern.

Two mechanisms remove the *counted* premium when added alone: private valuations (-0.036) and ministerial policy (-0.029). In E* the counted premium is small ($+0.052$), and it returns (0.21–0.22) when private valuations, heterogeneous valuations (without which private types carry no information) or exits are removed, but not when ministerial policy is removed. The mechanisms therefore interact: on its own, ministerial policy turns the premium into policy control (lead coefficient $+0.803$), but in E* without private valuations the formateur keeps both more ministries and more lead ministries (lead coefficient 0.35). We report these interactions as exploratory; with three seeds per configuration we did not decompose them further. But the premium does not vanish. In each party's share of the value it places on ministries the formateur coefficient is $+0.107$ with private valuations and $+0.205$ in E*, and with ministerial policy the formateur holds the policy-setting ministries (coefficient $+0.803$). Both persist with four times the training. The formateur gives partners more ministries but keeps the ones it values most, or

Table 2: Formateur coefficient $\hat{\gamma}$ of regression (2) by configuration (mean ± standard error over seeds). Counted: share of ministries. Value: share of the value each party places on ministries. Lead: share of policy-setting ministries (only with M9). $\tilde{D}_G$: floor-adjusted Gamson distance. Configurations with 2–3 seeds are exploratory; the confirmatory tests are in Section 5.3.

configuration	$\hat{\gamma}$ counted	$\hat{\gamma}$ value	$\hat{\gamma}$ lead	$\tilde{D}_G$	seeds
base game (all mechanisms off)	+0.249 ± 0.014	–	–	0.132	5
+ repeated terms (M1)	+0.291 ± 0.014	–	–	0.154	3
+ repeated, patient	+0.320 ± 0.010	–	–	0.182	3
+ heterogeneous public values (M5)	+0.172 ± 0.034	+0.131 ± 0.026	+0.227 ± 0.037	0.149	3
+ private values (M4+M5)	−0.036 ± 0.003	+0.107 ± 0.004	−0.032 ± 0.009	0.072	5
+ exits (M8)	+0.223 ± 0.028	–	–	0.138	3
+ ministerial policy (M9)	−0.029 ± 0.007	−0.029 ± 0.007	+0.803 ± 0.022	0.089	5
+ seat recognition (M10)	+0.159 ± 0.017	–	–	0.098	5
E* (all on)	+0.052 ± 0.015	+0.205 ± 0.016	+0.156 ± 0.025	0.093	10
E* − repeated terms	−0.040 ± 0.014	+0.105 ± 0.014	−0.004 ± 0.014	0.098	3
E* − private values	+0.221 ± 0.009	+0.200 ± 0.009	+0.348 ± 0.023	0.108	3
E* − heterog. values	+0.210 ± 0.014	+0.210 ± 0.014	+0.203 ± 0.132	0.090	3
E* − exits	+0.207 ± 0.033	+0.340 ± 0.032	+0.252 ± 0.036	0.097	3
E* − ministerial policy	+0.053 ± 0.020	+0.211 ± 0.019	+0.040 ± 0.017	0.093	3
E* − seat recognition	+0.092 ± 0.021	+0.244 ± 0.021	+0.168 ± 0.024	0.116	3
E*, MAPPO	+0.037 ± 0.013	+0.188 ± 0.013	+0.124 ± 0.022	0.085	10
E*, fictitious self-play	+0.065 ± 0.028	+0.219 ± 0.029	+0.153 ± 0.056	0.094	3
E*, GRU memory	+0.040 ± 0.029	+0.196 ± 0.029	+0.185 ± 0.060	0.097	3
E*, memoryless	−0.006 ± 0.009	+0.142 ± 0.009	+0.069 ± 0.043	0.096	3
E*, 12 ministries	−0.059 ± 0.030	+0.147 ± 0.031	+0.548 ± 0.111	0.122	3
base, 4× training	+0.295 ± 0.004	–	–	0.158	2
+ private values, 4×	−0.004 ± 0.000	+0.139 ± 0.001	−0.028 ± 0.021	0.070	2
+ policy, 4×	+0.007 ± 0.021	+0.007 ± 0.021	+0.770 ± 0.077	0.085	2
E*, 4×	+0.088 ± 0.041	+0.240 ± 0.041	+0.158 ± 0.010	0.090	2
base, PSRO	+0.301	–	–	–	1
E*, PSRO	+0.070	–	–	–	1

keeps policy control: the premium changes currency.

5.3 Demand bargaining follows voting weights

On the same random 3–8-party parliaments (10 seeds per protocol), demand bargaining lowers the formateur coefficient by 0.34 relative to proposal bargaining (95% CI 0.32–0.36; H1) and raises the seat slope by 0.36 (0.34–0.37; H2). Each test is one-sided and exact: the unit is a training seed, each seed contributing one coefficient estimated on its own evaluation cabinets; the 20 seed-level estimates are pooled and the protocol labels are permuted over all $\binom{20}{10} = 184{,}756$ assignments. Because the two protocols' estimates do not overlap, the observed difference is the most extreme assignment and $p = 1/184{,}756 \approx 5 \times 10^{-6}$, the smallest value the test can give; the intervals are percentile bootstraps over seeds. The demand coefficient meets the pre-registered practical-equivalence criterion $|\gamma| < 0.05$ (−0.039, 90% CI −0.047 to −0.030; H3, two one-sided t-tests). This is equivalence within the margin, not evidence of an exact zero: the interval lies slightly below zero. All three tests pass after Holm correction. The small negative value is an exploration artefact: with five times the entropy on the demand head the coefficient becomes −0.009 to +0.003 (seat slope 0.82–0.86). We have not isolated why. Raising entropy does not make closing more likely a priori (one close action among 102), but it keeps demands from collapsing early onto a few values, so later movers keep meeting undercutting demands and cheap closing options.

Is this Gamson's law? The fixed parliaments of Table 3 separate seats from voting weights. Learned demand agents reproduce the exact equilibrium of Section 3.2: every 45/35/20 pair splits within 0.02 of 50/50 (pre-registered prediction P1: every seed's slope below 0.5; seat slope

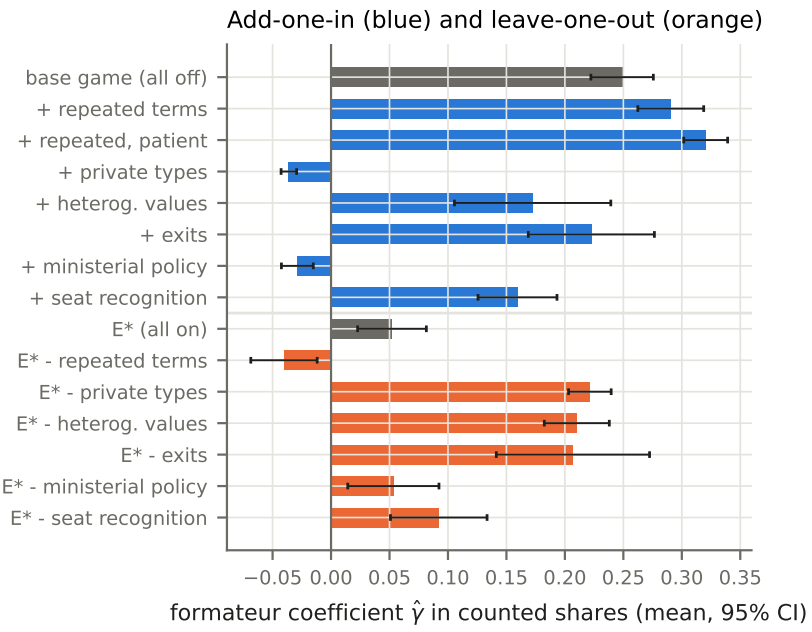


Figure 4: Formateur coefficient $\hat{\gamma}$ in counted shares when a mechanism is added to the base game (blue) or removed from E* (orange); mean and 95% CI over seeds.

Table 3: Who gets what in fixed parliaments (divisible surplus, $\delta = 0.9$, seat-proportional recognition; learned: 3 seeds per protocol, every coalition that forms).

	45/35/20 (each pair)	40/30/25/5
Gamson's law (seats)	larger party 0.56–0.69	dummy gets $5/w(C)$ if included
demand: exact equilibrium	0.50/0.50	0.50/0.50, dummy never included
demand: learned	0.50 ± 0.02 (seat slope -0.02 to $+0.12$)	0.50 ± 0.03, dummy included 0.0%
proposal: weighted BF	formateur ≈ 0.70	large-party formateur 0.71; dummy formateur 0.41
proposal: learned	formateur 0.65–0.73	large-party formateur 0.65–0.71; dummy 0.36–0.39

-0.02 to $+0.12$, seed-level 95% CI -0.12 to $+0.24$, which excludes Gamson's slope of 1), and the dummy party in 40/30/25/5 is never included (P2). Learned proposal agents give the formateur 0.65–0.73, close to the weighted Baron–Ferejohn solution (P3). In 40/30/25/5 the 5-seat party is recognised in about 5% of rounds (recognition is seat-proportional); it must then buy two large partners and keeps 0.36–0.39, against 0.41 in the weighted Baron–Ferejohn solution. Learned demand shares thus follow voting weights; the seat slope of 0.8 in random parliaments arises because seats and voting weights are correlated there. In symmetric demand games with default exploration the closer is under-paid (0.35 with three parties, 0.16 with five) and a best response gains +0.14 per period; with more exploration of demands the closer gets 0.44 and 0.34, moving to the equilibrium values 0.50 and about 0.34, and the best-response gain falls to 0.029.

5.4 Real cabinets

We use ministers per party from the Seki–Williams party government data (Seki and Williams, 2014), linked to ParlGov parliaments (Döring and Manow, 2026): 195 majority coalition cabinets in 31 countries, 1988–2014, with the prime minister's party as the formateur. Real cabinets were never used for training or tuning. *Samples.* All regressions and robustness checks use the 195 cabinets. Split predictions use 183: the agents' parliaments contain the eight largest parties,

Table 4: Real cabinets: the prime-minister coefficient $\hat{\gamma}$ of (2) under different specifications, with wild-cluster bootstrap 95% CIs (31 countries). Bottom: partners with equal minimal integer voting weight but different seats (`scripts/real_weights_vs_seats.py`).

specification	$\hat{\gamma}$	95% CI	R^2
seat shares	+0.011	−0.023 to +0.045	0.93
seat shares, intercept α/k (cabinet size)	+0.019	−0.014 to +0.054	0.93
seat shares + largest-party dummy	+0.059	+0.022 to +0.096	0.94
minimal integer voting weights	+0.074	+0.025 to +0.128	0.88
seats and weights together	+0.011	−0.024 to +0.045	0.94
equal voting weight, different seats: seat slope 0.84 (95% CI 0.44–1.07), 42 pairs; without the PM's party 0.52 (24 pairs)			

which excludes the 12 cabinets with a member outside them. The key-ministry test uses the 131 cabinets (25 countries) that can be matched to the Seki–Williams minister records with at least four partisan senior ministers, a prime minister's party identifiable among them, a partner holding a ministry, and at least one key ministry. The equal-weight test uses the 42 partner pairs with equal minimal integer weight and seat shares differing by more than 0.02. Coalition choice uses ParlGov's 242 Western European parliaments.

The real regression. Real cabinets follow Gamson's law closely: share $= 0.031 + 0.896\,s + 0.011\,F$, $R^2 = 0.93$; leaving out any one country keeps $\hat{\gamma}$ between -0.004 and $+0.013$. The formateur premium depends on the measure of strength (Table 4): none relative to seats, +0.074 relative to minimal integer voting weights, as Ansolabehere et al. (2005) report, and +0.059 with a largest-party control (identified by the 20 cabinets whose prime minister does not come from the largest partner). Seats fit better than weights and absorb the weight effect when both enter. The most direct comparison is between partners with *equal* voting weight but different seats, where demand theory and our learned demand agents predict equal shares and Gamson's law predicts seat-proportional ones. For every pair of members i, j of cabinet c with equal minimal integer weight and seat shares differing by more than 0.02 we estimate, without intercept,

$$p_{ic} - p_{jc} = \beta\,(s_{ic} - s_{jc}) + \gamma\,(F_{ic} - F_{jc}) + \varepsilon_{ijc}, \tag{3}$$

so that equal seats and no prime minister imply equal shares. (Without an intercept the estimate does not depend on which partner of a pair is labelled i: swapping them changes the sign of every term.) Gamson's law predicts $\beta = 1$; voting-weight theories, and our learned demand agents, predict $\beta = 0$. We find $\hat{\beta} = 0.84$ with a 95% interval of 0.44–1.07 from a bootstrap over countries, because pairs from the same cabinet or country are not independent. Without the pairs that involve the prime minister's party, $\hat{\beta} = 0.52$ (0.28–1.15; 24 pairs from only 7 countries). Both intervals exclude zero, but they are wide: the estimate supports seat-based division without measuring it precisely.

Key ministries. If the real premium were hidden in quality, as with our agents, prime ministers' parties should hold more than their share of the key ministries. On 131 cabinets they hold 0.589 of finance, foreign affairs, interior, defence and justice, against 0.573 of all ministers (difference +0.015, 95% CI −0.021 to +0.056): no detectable premium, in line with salience-weighted evidence (Warwick and Druckman, 2006).

Prediction. To predict the split of a real cabinet, the agent of the prime minister's party acts as formateur in the real parliament (its eight largest parties) with the real number of ministries; the coalition-inclusion steps of its proposal head are fixed to the real coalition, and the ministry pointer head is sampled 32 times and averaged. Demand agents are not scored this way: fixing

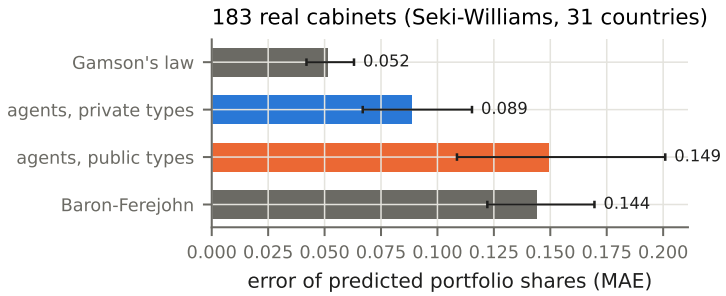


Figure 5: Predicting the portfolio split of real coalition cabinets given the real coalition and prime minister: mean absolute error per party with country-clustered 95% CIs.

Table 5: Regression (2) with seat shares and with Banzhaf voting power ($\hat{\gamma}_{bz}$) on real cabinets and on the cabinets that agents form themselves in the same 195 parliaments (ranges over seeds: 10 per protocol, 3 otherwise).

	$\hat{\beta}$	$\hat{\gamma}$	$\hat{\gamma}_{bz}$	R^2
real cabinets	0.896	0.011	+0.056	0.93
demand protocol	0.74–0.87	−0.07 to −0.01	−0.08 to −0.02	0.80–0.89
with more exploration	0.86–0.87	−0.02 to 0.00	−0.03 to 0.00	0.86–0.87
proposal protocol	0.49–0.55	+0.21–0.34	+0.24–0.38	0.84–0.94
E* (proposal, all mechanisms)	0.63–0.67	+0.01–0.08	+0.05–0.11	0.71–0.73

the coalition removes the outside options they were trained with (their demands then exceed the surplus in 12–22% of cabinets, depending on the seed), so we evaluate them on the cabinets they form themselves (Table 5). Given each real coalition and prime minister, proposal policies trained only on random parliaments predict the split with mean absolute error 0.087–0.090 per party when they have private valuations, and 0.130–0.167 with public valuations; Baron–Ferejohn errs by 0.144 and Gamson's law by 0.052: Gamson's law has the lowest error of the models scored on this common fixed-coalition task, in every leave-one-country-out sample (Figure 5). Demand agents are compared with real cabinets only through the cabinets they form themselves (Table 5), so this ranking does not include them.

Agents in real parliaments. Left to bargain freely in the real parliaments (Table 5), demand agents with more exploration reproduce the real seat regression almost exactly, although their rule is voting weight: in most real parliaments the two are close. E* agents match the small premium relative to voting power; proposal agents overshoot it. Restricting the agents' cabinets to those that reproduce the real coalition barely changes the demand slope (0.68–0.83), so coalition selection is not what makes it seat-like. Choosing coalitions in 242 Western European parliaments, the agents pick the governing coalition in 21.7% of cases, similar to Riker's minimal-seat rule (19.7%) and below an ideologically connected rule (31.7%).

5.5 Reliability

Table 6 reports estimated exploitability. Best responses gain +0.015 per period against E* (10% of a seat's payoff) and about a fifth of a seat's payoff against agents on random parliaments under either protocol, against +0.037 and +0.088 for scripted Gamson and Baron–Ferejohn-heuristic profiles. The test has power at 300–400 iterations (it finds the gaps of Section 5.1), but at 150 iterations its controls in the base game were not detected, so budgets matter. Cross-play between independently trained runs costs +0.0020 per period within an algorithm and +0.0004 across algorithms: independent runs settle on compatible conventions. Learned profiles are approximate equilibria, and our conclusions are about the allocation rules they settle on, which are stable

Table 6: Estimated exploitability: per-period gain of a best response trained by PPO for one random seat against the frozen profile (larger of a warm and a from-scratch start), also relative to the profile seat's payoff. Scripted positive controls (proposal protocol only) have no network weights to start from, so their best response is trained from scratch only; with a weak budget its gain can be slightly negative.

profile	BR iters	gain	relative	Gamson ctrl.	BF-heur. ctrl.
$E^\star$ (IPPO)	400	0.015	10%	0.037	0.088
$E^\star$ (MAPPO)	150	0.018	12%	0.023	0.094
$E^\star$ (IPPO, 3 seeds)	150	0.016	10%	0.020	0.076
base game (3 seeds)	150	0.017	7%	0.008	−0.004
E0 (BF benchmark)	300	0.040	20%	0.019	0.037
random parliaments, proposal	400	0.033	18%	–	–
random parliaments, demand	400	0.034	19%	–	–
symmetric demand, default exploration	300	0.143	57%	–	–
symmetric demand, more exploration	300	0.029	12%	–	–

across seeds, learners and training budgets.

6 Discussion

Answer to the question. Should cabinets allocate proportionally or favour formateurs? Read positively, as a question about what self-interested bargaining produces, our answer is: neither rule emerges unconditionally. Learning parties favour the formateur exactly as far as the rules give it proposal power (Section 5.2); when parties bargain by demands, the premium disappears but shares follow voting weights (Section 5.3). Private information and policy motives do not remove the proposer's advantage; they move it into the ministries the formateur values or into policy control, which counting ministries cannot see. Real cabinets, in contrast, divide by seats even where voting weights disagree, show at most a small premium relative to voting weights, and no detectable premium in key ministries (Section 5.4).

Implications. Gamson's law is therefore not the equilibrium of either canonical bargaining protocol, as learned by self-interested agents. Demand bargaining comes closest, and in real parliaments, where seats and voting weights are correlated, its learned allocations reproduce the real regression. Where seats and weights diverge, the available evidence has real cabinets siding with seats. This points to mechanisms outside these bargaining models: commitments to a coalition before the division (Kamm and Siegenthaler, 2024), pre-electoral pacts (Carroll and Cox, 2007), intra-party bargaining (Cox, 2021) or a proportionality norm. We do not claim that real governments bargain by our demand protocol, or that its equilibrium describes demand bargaining in general; it is one well-defined alternative to proposal power, solved exactly. For the learning literature, the benchmark shows how outcome-level validation can hide non-equilibrium play: the learned profile matches Baron–Ferejohn shares while leaving a profitable deviation untried.

Limitations. Sizes are scaled down ($n \leq 5$ in the mechanism experiments, 3–8 in random and real parliaments, $m = 6$). Learned profiles are approximate equilibria, and exploitability estimates are lower bounds. The prime minister's party is a proxy for the formateur. Real shares count ministers; salience enters only through the key-ministry test, and the test of the currency shift in real data with party-specific manifesto salience was not run. The demand head offers one close action against 101 demand bins, which slows learning to close; the exploration runs show that our conclusions do not depend on it. The normative comparison of allocation conventions (delay, durability, welfare) was outside the scope of the experiments reported here.

7 Conclusion

Modelling parties as learning agents bargaining over cabinets shows that the formateur premium is a property of proposal power, not of strategic bargaining as such, and that its absence under demand bargaining yields division by voting weight, not by seats. Information and policy motives change where the premium appears rather than whether it exists. Where seats and voting weights disagree, real cabinets follow seats, unlike any protocol we trained; the evidence for this rests on 42 partner pairs and its interval is wide, but it suggests that the explanation of Gamson's law must include rules or norms beyond the canonical bargaining models.

Reproducibility. Code, configurations, trained agents, raw results, the pre-registration and a map from every claim to its result file and script are at `https://github.com/bot2048/cabinet-bargaining-marl` (`docs/RESULTS_AUDIT.md`). Every number in this paper is generated from the result files (`scripts/make_figures.py`, `scripts/paper_tables.py`); 32 automated tests tie the code to the theory (`tests/test_core.py`). We re-ran every analysis and the test suite from a fresh clone of the repository; all outputs were identical to the committed ones.

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A Proof of Proposition 1

Embedding. Fix a stationary profile at the divisible level with a cabinet that governs forever, no exits and constant ρ. A member j of an accepted cabinet with share p_j receives $p_j/(1-\delta)$; after a rejection it gets nothing this period and δV_j afterwards, with $V_j = v_j/(1-\delta)$. Stage-undominated voting accepts if and only if $p_j \geq \delta v_j$. A proposal passes only if $C \ni f$, $w(C) \geq q$ and every partner receives at least δv_j, so the best passing proposal pays exactly δv_j to a cost-minimising coalition and leaves $x_f = 1 - \delta c_f(v)$, where $c_f(v)$ is the minimal $\sum_{j\in S} v_j$ over sets S with $w(S \cup \{f\}) \geq q$. Because the grand coalition is winning, $c_f(v) \leq 1 - v_f$ and $x_f \geq 1 - \delta + \delta v_f > \delta v_f$, so passing is optimal. These are exactly the conditions of the closed-rule game of Baron and Ferejohn (1989), so stationary strategies and per-period shares coincide. □

Symmetric case. Conjecture $v_i = 1/n$. Voters accept offers of at least δ/n. Every coalition of f and $(n-1)/2$ others costs $(n-1)/(2n)$, so f keeps $x_f^{\mathrm{BF}} = 1 - \delta(n-1)/(2n)$ and, choosing uniformly among cheapest coalitions, includes each other party with probability $1/2$. Then

$$v_i = \tfrac{1}{n}\Big(1 - \tfrac{\delta(n-1)}{2n}\Big) + \tfrac{n-1}{n} \cdot \tfrac{1}{2} \cdot \tfrac{\delta}{n} = \tfrac{1}{n},$$

confirming the conjecture. By the one-shot deviation principle the profile is a stationary equilibrium; stationary equilibrium payoffs are unique for quota rules (Eraslan, 2002). □

B The demand-protocol solver

For a move order π and a demand grid G, let $h = (d_1, \ldots, d_k)$ be the demands of the first k movers. The mover at position k can close if some set S of earlier movers satisfies $w(S) + w_{\pi_k} \geq q$ and $\sum_{j\in S} d_j \leq 1$; the environment then takes the cheapest such S (ties: fewer members, then a fixed subset order) and pays the closer $1 - \sum_{j\in S} d_j$. Otherwise, or by choice, it demands $a \in G$ and play moves to (h, a); at the last position a demand ends the round with payoffs δV. Backward induction over all $|G|^k$ histories, vectorised over histories, gives every mover's optimal action; the ex-ante values V are averaged over recognition and random orders and iterated to a fixed point. Indifferent movers break ties by a fixed rule (close first; then the highest or the lowest demand), and the reported equilibria are identical under both rules.

Verification (`scripts/demand_solver_check.py`). (i) An independent implementation, a plain recursive backward induction over one history at a time with its own cheapest-set routine, gives the same ex-ante values and the same on-path outcome for every move order in 14 configurations (three and five equal parties, 45/35/20, 40/30/25/5; grids of 10% and 5%, 10% only for five parties; both tie rules), up to single-precision rounding (10^{-8}). (ii) That cheapest-set routine selects the same coalition as the environment in all 2000 random demand histories with 3–6 parties (0 mismatches). (iii) The fixed-point iteration in V stops with residual zero to machine precision after one or two iterations: on the equilibrium path no round fails, so δV only bounds off-path options and never binds. (iv) The on-path splits are identical under both tie rules, every grid and $\delta \in \{0.5, 0.9, 0.95\}$. Automated tests cover the three-party and dummy-party cases and the agreement with the brute-force solver (`tests/test_core.py`).

C Hyper-parameters

Table 7: PPO hyper-parameters (`cabinet/ppo.py`, `DEFAULT_HP`), tuned on E0 only.

learning rate	$3 \cdot 10^{-4}$ (Adam)	decisions per iteration	8192 (64 environments)
GAE λ	0.95	PPO epochs, minibatches	4, 4
clip	0.2	value-loss weight	0.5
entropy bonus	$2 \cdot 10^{-2} \to 2 \cdot 10^{-3}$	entropy multiplier, vote/exit	5
embedding width	64, 4 heads, 2 blocks	demand-entropy multiplier	1 (5 with more exploration)

D Environment parameters

Table 8: Environment parameters (`cabinet/config.py`, `cabinet/env.py`).

parliament (base game, E*)	$n \in \{3, 4, 5\}$ uniform; seat shares $\sim$ Dirichlet(1.5) on a house of 100; quota $\lfloor W/2 \rfloor + 1$
parliament (random 3–8 parties)	$n \in \{3, \dots, 8\}$; Dirichlet concentration drawn from $\{0.5, 1, 2, 3\}$; house of 150; seat-proportional recognition
benchmark E0	equal seats, $n \in \{3, 5, 7\}$, $\delta \in \{0.8, 0.9, 0.95\}$, uniform recognition, divisible surplus
discount	$\delta = 0.9$ unless stated
ministries	$m = 6$, equal salience $\sigma_k = 1/m$
valuations (M5, M4)	$\nu_{ik} = (1 - \xi)\sigma_k + \xi\,\epsilon_{ik}$, $\xi = 0.5$, $\epsilon_{i\cdot} \sim$ Dirichlet$(\kappa_\nu/m, \dots)$ with $\kappa_\nu = 5$, rounded to 0.01; private under M4
policy (M9)	$d = 2$ dimensions; ideal points uniform on $[0, 1]^2$ (grid 0.05); status quo $y^0 = 0.5$; $\lambda = 0.5$; ministries 1 and 2 lead dimensions 1 and 2
terms (M1)	$L = 6$ periods per term; the game continues after each term with probability $\beta_T = 1 - 1/T$, $T = 3$; seats at an election $\sim$ Dirichlet$(\kappa_w \bar{\chi})$ around the initial shares $\bar{\chi}$, $\kappa_w = 200$; memory of the last $\Lambda = 4$ cabinets
patient repetition	$L = 3$, $T = 6$, $\delta = 0.98$
exits (M8)	each member's exit bonus $\bar{e} \cdot \text{Exp}(1)$, $\bar{e} = 0.05$, rounded to $\bar{e}/20$ and capped at $10\bar{e}$, drawn privately each governed period
one-shot games	a cabinet governs forever; shares are paid once as per-period equivalents (embedding of Section A); formation stops after 40 failed rounds, which matters neither on nor off the path: the discount weight of round 40 is at most $0.95^{40} < 0.13$, and with per-round failure rates of at most 18% in any run the probability of reaching it is below 10^{-29}

E Learning curves

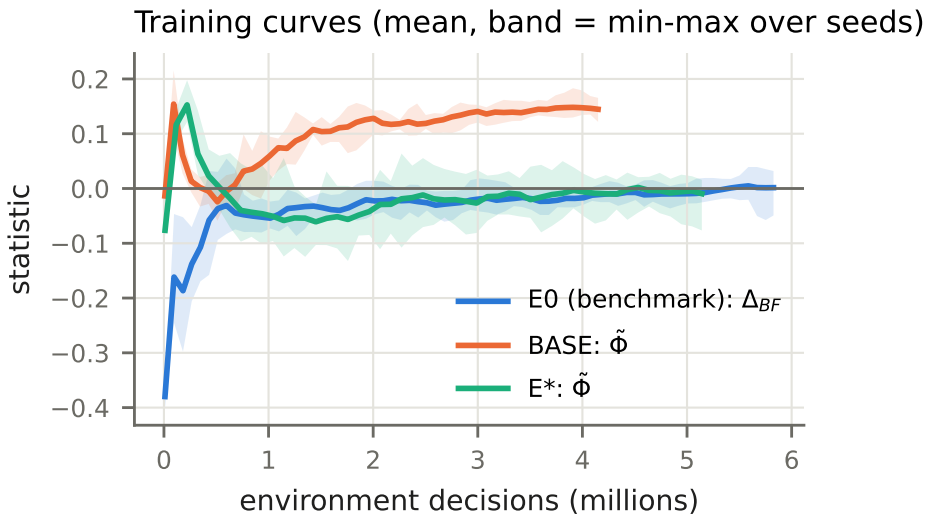


Figure 6: Training curves (mean over seeds; band: minimum to maximum). E0: gap of the learned formateur share to the closed form. Base game and E*: floor-adjusted formateur premium $\tilde{\Phi} = \mathbb{E}[p_f - p_f^{\min}]$, where $p^{\min}$ is the most proportional feasible allocation of the same cabinet. The premium of the base game is absent early in training and grows with it.